

NMSM Homework Exercises

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Sampling

1.1 Sampling in d -dimensional domains by hit-and-miss

Source codes `src/011a_rect_hit_miss.c, src/011b_disk_hit_miss.c`

Analysis scripts `src/011_hit_miss.R`

Executables `exe/011a_rect_hit_miss, exe/011b_disk_hit_miss`

The error as a function of the number of throws is shown in Fig. 1.1.1. In the disk case, it goes under 1 % with as little as 10 000 ‘throws’.

1.2 Sampling from a distribution with the inversion method

Source codes `src/012_inversion_power34.c`

Analysis scripts `src/012-4_inversion_method.R`

Executables `exe/012_inversion_power34`

The idea is to sample from the probability distribution $\rho_n(x) = cx^n$ in $[0, 1]$. First, using the normalization condition we can find out what c should be:

$$1 = \int_0^1 cx^n dx = \frac{c}{n+1} \implies c = n+1. \quad (1.2.1)$$

Then, we find the expression of the associated cumulative density function:

$$F_n(x) = (n+1) \int_0^x y^n dy = x^{n+1}, \quad (1.2.2)$$

and invert it:

$$u = x^{n+1} \implies x = u^{1/(n+1)}. \quad (1.2.3)$$

So, we just have to sample a random floating point number between 0 and 1 and raise it to the power of $1/(n+1)$ to get $x \sim (n+1)x^n$. Results from a sampling of 50 000 iterations for $n = 3, 4$ are shown in Fig. 1.2.1.

1.3 Another sampling with the inversion method

Source codes `src/012_inversion_power2.c`

Analysis scripts `src/012-4_inversion_method.R`

Executables `exe/012_inversion_power2`

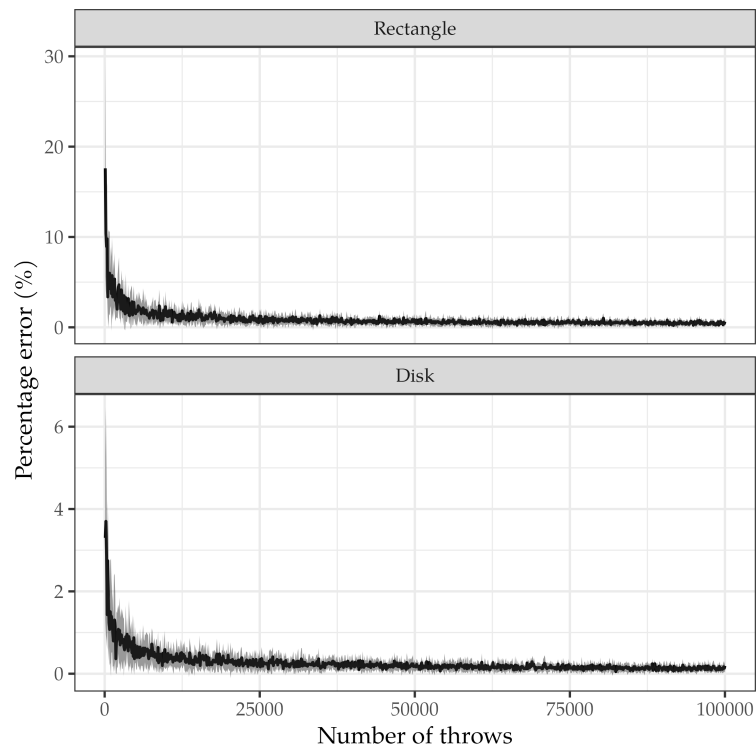


Figure 1.1.1: percentage error as a function of the number of ‘throws’ for Ex. 1.1. The black line shows the average over 10 realizations, with the standard deviation shaded in grey.

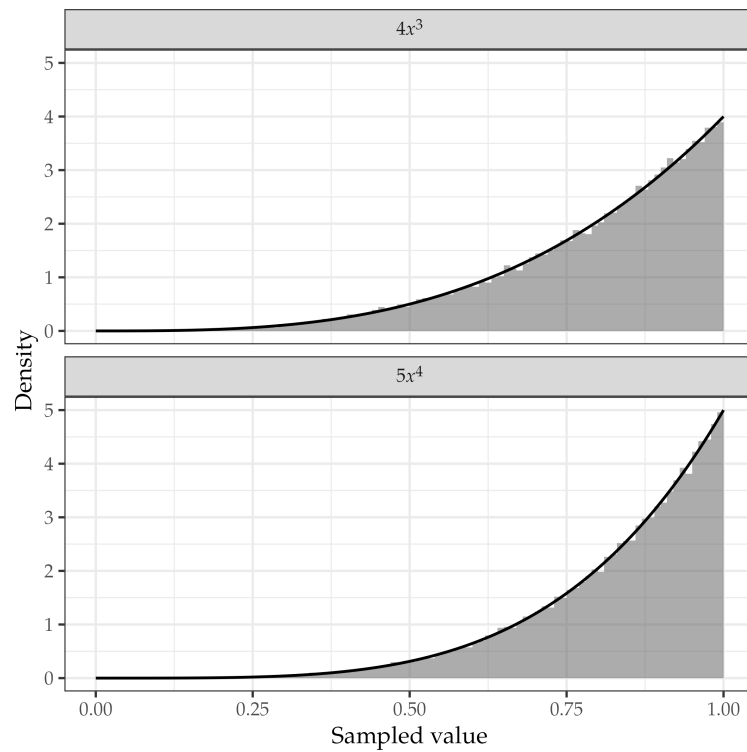


Figure 1.2.1: histogram of 50 000 samples from $\rho_n(x) = cx^n$, $x \in [0, 1]$, for $n = 3$ (up) and $n = 4$ (down), obtained with the inversion method for Ex. 1.2. The theoretical distribution is overlaid with a solid line.

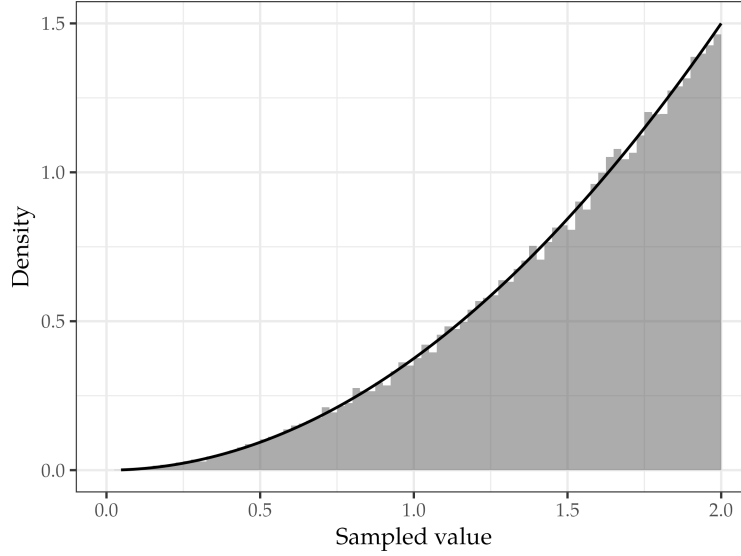


Figure 1.3.1: histogram of 50 000 samples from $\rho(x) = 3x^2/8$, $x \in [0, 2]$, obtained with the inversion method for Ex. 1.3. The theoretical distribution is overlaid with a solid line.

I slightly modified the code from the previous exercise, since c is different now. The distribution is $\rho(x) = cx^2$ with $x \in [0, 2]$, so:

$$1 = \int_0^2 cx^2 dx = \frac{8}{3}c \implies c = \frac{3}{8}. \quad (1.3.1)$$

The cumulative is then

$$F(x) = \frac{3}{8} \int_0^x y^2 dy = \frac{x^3}{8} \implies x = 2u^{1/3}. \quad (1.3.2)$$

Once again, you can see the comparison between a 50 000-points histogram and the theoretical curve in Fig. 1.3.1.

1.4 Additional exercises on the inversion method

Source codes `src/014a_inversion_exp.c`, `src/014b_inversion_exp2.c`,
 `src/014c_inversion_powerlaw.c`

Analysis scripts `src/012-4_inversion_method.R`

Executables `exe/012_inversion_exp`, `exe/014b_inversion_exp2`,
 `exe/014c_inversion_powerlaw`

The first exercise here asked to sample from $\rho_1(x) = \mu e^{-\mu x}$, with $x \in [0, \infty)$. The cumulative and the sampling rule are then

$$F_1(x) = \mu \int_0^x e^{-\mu y} dy = \mu(1 - e^{-\mu x}) \implies x = -\log(1 - u)/\mu. \quad (1.4.1)$$

I ran 50 000 samples as before, choosing $\mu = 2$; see Fig. 1.4.1.

The second distribution was $\rho_2(x) = 2xe^{-x^2}$, $x \in [0, \infty)$. Cumulative and sampling rule:

$$F_2(x) = 2 \int_0^x e^{-y^2} dy = 1 - e^{-x^2} \implies x = \sqrt{-\log(1 - u)}. \quad (1.4.2)$$

A histogram of 50 000 samples is shown in Fig. 1.4.2.

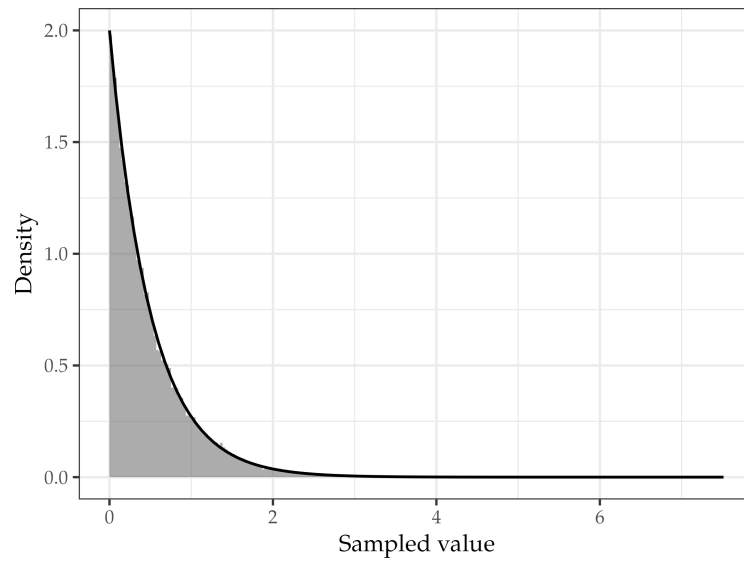


Figure 1.4.1: histogram of 50 000 samples from $\rho(x) = 2e^{-2x}$, $x \in [0, \infty)$, obtained with the inversion method for the first part of Ex. 1.4. The theoretical distribution is overlaid with a solid line.

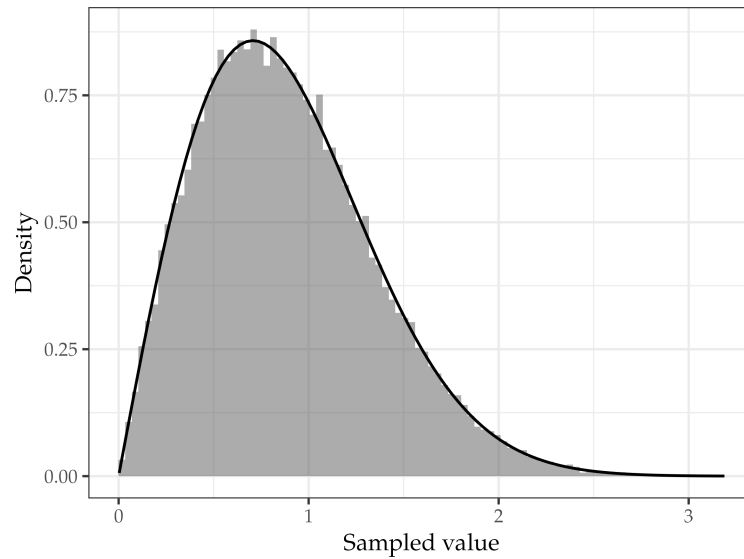


Figure 1.4.2: histogram of 50 000 samples from $\rho(x) = 2xe^{-x^2}$, $x \in [0, \infty)$, obtained with the inversion method for the second part of Ex. 1.4. The theoretical distribution is overlaid with a solid line.

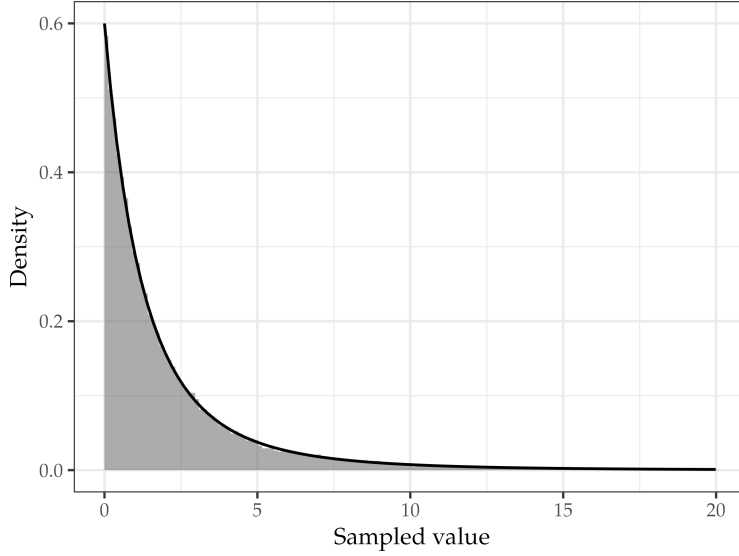


Figure 1.4.3: histogram of 50 000 samples from $\rho(x) = 375/(5+x)^4$, $x \in [0, \infty)$, obtained with the inversion method for the third part of Ex. 1.4. The theoretical distribution is overlaid with a solid line.

The last distribution was $\rho_3(x) = c(a+bx)^{-n}$, with $x \in [0, \infty)$ and $n > 1$. The normalization constant is given by

$$1 = \int_0^\infty c(a+bx)^{-n} dx = \frac{a^{1-n}c}{b(n-1)} \implies c = b(n-1)a^{n-1}. \quad (1.4.3)$$

By integrating we obtain

$$F_3(x) = c \int_0^x (a+by)^{-n} dy = \frac{c}{b(1-n)} [(a+bx)^{1-n} - a^{1-n}] = 1 - a^{n-1}(a+bx)^{1-n}, \quad (1.4.4)$$

from which we can sample with

$$u = 1 - a^{n-1}(a+bx)^{1-n} \implies x = \frac{a}{b} [(1-u)^{1/(1-n)} - 1]. \quad (1.4.5)$$

I chose $a = 1/2$, $b = 1/10$ and $n = 4$ – resulting in $\rho_3(x) = 375/(5+x)^4$ – for a run of 50 000 samples, shown in Fig. 1.4.3.